

Help for command OriCreateCalib

Contents

1	Description of OriCreateCalib	1
2	Command related to OriCreateCalib	2

1 Description of OriCreateCalib

This command create initial intrinsic calibration for in MMVIIformat for a set of images. It take 2 mandatory parameter, a set of images and destination folder for the generated model of calibration. For each image :

- the name of calibration file is computed (the same as can be seen in `ImageMetadata`);
- an initial calibration is created and saved using the folder and the name of the file.

The initial calibration is computed using the following approach :

- the focal length is extracted from meta-data, this requires that the focal in mm and the model of camera must be known (it works also if only the focal equiv 35 is known)
- the principal point is set in the middle of the image;
- the model of projection is set by optional parameter `Proj`, it's default value `Stenope` correspond to a rectilinear model, othe value correspond to different fisheyes or to 360 camera;
- the distorsion is initially void, it's degree of parametrisation is set by optionnal parameter `Degree`.

2 Specifying the degree of polynomial

Regarding the degree $D = [D_x, D_y, D_z]$:

- D_x fix the number of parameter for radial distorsion $D_x = 3$ will generate R^3, R^5, R^7 ;
- D_y fix the number of pameter fo decentric distortion (practically D_y is always equal to 1 or 0);
- D_z fix the degree of complementary polynom, the current value $D_z = 1$ correspond to affine function.

The default value for D is $[3, 1, 1]$ which correspond to the Fraser's model. Only some possible value are accepted for the degree, the value for which code have been generated. This possible value depend of `Proj`. To know what are the possible value, run `OriCreateCalib` with $D_x = -1$.

3 Command related to OriCreateCalib

See also :

- `EditCalcMTDI`
- `ImageMetadata`
- `OriPoseEstimSpaceResection`